

Time Series Exercise Sheet 5

Sofia Olhede

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Exercise 5.1

Suppose that $\{X_t\}$ is an ARIMA(p, d, q) process satisfying the equations

$$\phi(B)(I - B)^d X_t = \theta(B)Z_t$$

where Z_t is white noise. Show that the same equations are also satisfied by $W_t = X_t + A_0 + A_1 t + \dots + A_{d-1} t^{d-1}$, where A_j are independent random variables.

Solution 5.1

Firstly, if we apply the differencing operator to the zero sequence (i.e. $0 = \{0\}_{t \in \mathbb{Z}}$) then we have

$$(I - B)0 = 0.$$

Furthermore, since the differencing operator is linear, we can consider the effect on each part of term in W_t , in other words

$$(I - B)^d W_t = (I - B)^d X_t + A_0(I - B)^d 1 + A_1(I - B)^d t + \dots + A_{d-1}(I - B)^d t^{d-1}.$$

Combining these two observations, we see that we need only show that

$$(I - B)^k t^{k-1} = 0$$

for all $k \in \mathbb{N}$.

The proof of this can be achieved by induction. If $k = 1$ then we have

$$(I - B)t^0 = (I - B)1 = 0.$$

(Note here 0 and 1 refer to the zero and one sequences!)

Now assume that the statement holds for all $r \leq k$ for some k . Then we have

$$(I - B)^k t^{k-1} = 0.$$

Now we have

$$\begin{aligned} (I - B)^{k+1} t^k &= (I - B)^k (I - B) t^k \\ &= (I - B)^k (t^k - (t - 1)^k) \\ &= -(I - B)^k \sum_{r=0}^{k-1} \binom{k}{r} t^r (-1)^{r-k} \end{aligned}$$

because the term involving t^k cancels out. Now by linearity, the induction hypothesis and that $(I - B)0 = 0$, the result holds for $k + 1$. Therefore by induction, the result holds for all $k \in \mathbb{N}$.

Exercise 5.2

You observe U_t that follows a SARMA $(1, 2) \times (0, 1)_4$ model. Write down the equations that specify U_t in terms of the past. Determine the weights of the process in its infinite MA representation

$$U_t = \sum_{j=0}^{\infty} \psi_j \epsilon_{t-j}.$$

You may find it useful to define ψ_j in terms of ψ_{j-1} as this will simplify algebra.

Solution 5.2

From the lecture we rewrite the given SARMA model as:

$$\begin{aligned} (1 - \phi B)X_t &= (1 - \theta_1 B - \theta_2 B^2)(1 - \Theta B^4)\epsilon_t \Leftrightarrow \\ X_t &= \phi X_{t-1} + (1 - \theta_1 B - \theta_2 B^2)(1 - \Theta B^4)\epsilon_t. \end{aligned}$$

Recursively replacing X_{t-1} we may rewrite the above via an infinite series representation:

$$\begin{aligned} X_t &= \sum_{l=0}^{\infty} (\phi B)^l (1 - \theta_1 B - \theta_2 B^2)(1 - \Theta B^4)\epsilon_t \\ &= \sum_{l=0}^{\infty} (\phi^l B^l - \phi^l \theta_1 B^{l+1} - \phi^l \theta_2 B^{l+2} - \Theta \{\phi^l B^{l+4} - \phi^l \theta_1 B^{l+5} - \phi^l \theta_2 B^{l+6}\})\epsilon_t \\ &= \sum_{l=0}^{\infty} \psi_l \epsilon_{t-l}. \end{aligned} \tag{1}$$

We start now finding the coefficients ψ_l corresponding to the first few terms of ϵ_{t-l} .

Clearly $\psi_0 = 1$. Similarly we find the following:

$$\psi_1 = -\theta_1 + \phi; \quad \psi_2 = -\theta_2 - \theta_1 \phi + \phi^2; \quad \psi_3 = -\theta_2 \phi - \theta_1 \phi^2 + \phi^3 = \phi \psi_2.$$

Some care must be taken starting from ϵ_{t-4} as it is the first seasonal lag. Namely, proceeding similar to before, we compute:

$$\begin{aligned} \psi_4 &= -\Theta - \theta_2 \phi^2 - \theta_1 \phi^3 + \phi^4 = -\Theta + \phi \psi_3, \\ \psi_5 &= -\Theta(\phi - \theta_1) - \theta_2 \phi^3 - \theta_1 \phi^4 + \phi^5 \\ \psi_6 &= -\Theta(\phi^2 - \theta_1 \phi - \theta_2) - \theta_2 \phi^4 - \theta_1 \phi^5 + \phi^6 = -\Theta(\phi^2 - \theta_1 \phi - \theta_2) + (\phi^6 - \theta_1 \phi^5 - \theta_2 \phi^4) \\ \psi_7 &= -\Theta(\phi^3 - \theta_1 \phi^2 - \theta_2 \phi) + (\phi^7 - \theta_1 \phi^6 - \theta_2 \phi^5) = \phi \psi_6. \end{aligned}$$

Overall, we see a pattern arising from the recursion, which tells us that, for $l \geq 6$, ψ_l takes the form:

$$\psi_l = (\phi^l - \theta_1 \phi^{l-1} - \theta_2 \phi^{l-2}) - \Theta(\phi^{l-4} - \theta_1 \phi^{l-5} - \theta_2 \phi^{l-6}).$$

Exercise 5.3

Let us study the seasonal process $(1 - 0.7B^2)X_t = (1 - 0.3B^2)\epsilon_t$ for ϵ_t white noise with unit variance.

1. Find the coefficients $\{a_j\}$ in the representation $X_t = \sum_{j=0}^{\infty} a_j \epsilon_{t-j}$
2. Find the coefficients $\{b_j\}$ in the representation $\epsilon_t = \sum_{j=0}^{\infty} b_j X_{t-j}$
3. Find and visualise the auto-correlation of X_t .

Solution 5.3

1. We are given $(1 - \phi B^2) X_t = (1 - \theta B^2) \epsilon_t$, where $\phi = 0.7$ and $\theta = 0.3$. As the polynomial $\Phi(z) = (1 - \phi z^2)$ has no roots on the unit circle, X_t is stationary, and we may proceed as follows:

$$X_t = \frac{(1 - \theta B^2)}{(1 - \phi B^2)} \epsilon_t = \sum_{j=0}^{\infty} (\phi B^2)^j (1 - \theta B^2) \epsilon_t \quad (2)$$

Similar to the previous exercise, we compute the coefficients a_j corresponding to ϵ_{t-j} :

$$a_0 = 1; a_1 = 0; a_2 = -\theta + \phi; a_3 = 0; a_4 = -\phi\theta + \phi^2$$

Clearly, we see from eq. (2) that $a_j = 0$ for all odd j -s. Regarding j even we see that for $j \geq 2$ and even we have:

$$a_j = \phi^{j/2} - \theta^{\frac{j-2}{2}} \phi.$$

2. Similar to point a) we derive also in this case the representation:

$$\epsilon_t = \frac{(1 - \phi B^2)}{(1 - \theta B^2)} X_t = \sum_{j=0}^{\infty} (\theta B^2)^j (1 - \phi B^2) X_t,$$

from where it follows that $b_0 = 1$ and $b_j = 0$ for j odd. Then for $j \geq 2$ and even we compute once again:

$$b_j = \theta^{j/2} - \theta^{\frac{j-2}{2}} \phi.$$

3. We want to compute ρ_τ . We proceed as follows:

$$\begin{aligned} \text{Cov}(X_t, X_{t+\tau}) &= \text{Cov} \left(\sum_{j=0}^{\infty} a_j \epsilon_{t-j}, \sum_{k=0}^{\infty} a_k \epsilon_{t+\tau-k} \right) \\ &\stackrel{\text{Bounded Convergence}}{=} \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} a_j a_k \text{Cov}(\epsilon_{t-j}, \epsilon_{t+\tau-k}) \\ &\stackrel{\epsilon \text{ is white noise}}{=} \sigma_\epsilon^2 \sum_{j=0}^{\infty} a_j a_{j+\tau} \end{aligned}$$

and $\rho_\tau = \frac{\sum_{j=0}^{\infty} a_j a_{j+\tau}}{\sum_{j=0}^{\infty} a_j^2}$. We also compute $\text{Var}(X_t) = \sigma_\epsilon^2 \left\{ 1 + (1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \frac{\phi^2}{1-\phi^2} \right\}$.

Clearly, for τ being odd, we see that $\rho_\tau = 0$, $\rho_0 = 1$, while for $\tau \geq 2$ and even we compute:

$$\begin{aligned} \rho_\tau &= \frac{\sigma_\epsilon^2 \left\{ \phi^{\tau/2} - \phi^{\frac{\tau-2}{2}} \theta + \sum_{k=1}^{\infty} \left(\phi^{\frac{4k+\tau}{2}} - 2\phi^{\frac{4k+\tau-2}{2}} \theta + \phi^{\frac{4k+\tau-4}{2}} \theta^2 \right) \right\}}{\text{Var}(X_t)} \\ &= \frac{\sigma_\epsilon^2 \left\{ \phi^{\tau/2} - \phi^{\frac{\tau-2}{2}} \theta + \left(\phi^{\tau/2} - 2\phi^{\frac{\tau-2}{2}} \theta + \phi^{\frac{\tau-4}{2}} \theta^2 \right) \sum_{k=1}^{\infty} \phi^{2k} \right\}}{\text{Var}(X_t)} \\ &= \frac{\phi^{\tau/2} - \phi^{\frac{\tau-2}{2}} \theta + \frac{\left(\phi^{\tau/2} - 2\phi^{\frac{\tau-2}{2}} \theta + \phi^{\frac{\tau-4}{2}} \theta^2 \right) \phi^2}{1-\phi^2}}{\left\{ 1 + (1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \frac{\phi^2}{1-\phi^2} \right\}} \\ &= \phi^{\tau/2} \frac{1 - \phi^{-1}\theta + \frac{(1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \phi^2}{1-\phi^2}}{\left\{ 1 + (1 - 2\phi^{-1}\theta + \phi^{-2}\theta^2) \frac{\phi^2}{1-\phi^2} \right\}} \end{aligned}$$

The result is shown in Figure 1.

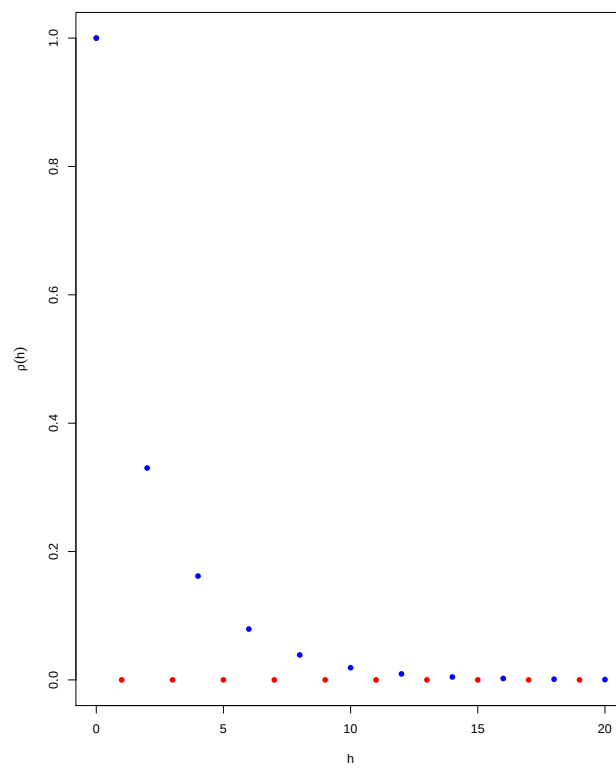


Figure 1: ρ_τ as a function of τ .